BALA MANEESH AYANALA

Full-Stack Developer · AI Engineer · IEEE Published

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Andhra Pradesh, India • Open to Bangalore, Hyderabad, Remote

PROFESSIONAL SUMMARY

Full-Stack Developer with hands-on experience building AI-powered web applications, real-time systems, and production-ready MERN platforms using React, Node.js, MongoDB, Docker, AWS, and LangChain. IEEE published researcher with experience in computer vision and deep learning using YOLOv8, PyTorch, and OpenCV. IBM SkillsBuild intern with proven delivery of dashboards, APIs, and CI/CD pipelines.

EDUCATION

B.Tech — Electronics & Communication Engineering *Sep 2021 – Jun 2025*

SRM Institute of Science and Technology (SRMIST), Chennai CGPA: 7.0/10

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL

Frontend: React.js, Tailwind CSS, Redux, Vanilla JS, HTML5, CSS3

Backend: Node.js, Express.js, FastAPI, REST APIs, WebSockets, JWT

Databases: MongoDB, PostgreSQL, ChromaDB

AI/ML: LangChain, RAG Pipelines, YOLOv8, PyTorch, OpenCV

Cloud/DevOps: Docker, AWS EC2, GitHub Actions, CI/CD, Netlify, Vercel

EXPERIENCE

Full-Stack Developer Intern — IBM Innovation Centre for Education (ICE) *Aug 2025 – Oct 2025*

Remote · IBM SkillsBuild

- Developed a MERN-stack automation dashboard centralising 15+ project metrics, improving reporting efficiency by 25%.
- Built reusable React components and integrated REST APIs, reducing manual operational effort by 30%.
- Deployed on AWS EC2 using Docker and GitHub Actions CI/CD pipelines, achieving 95+ Lighthouse scores.
- Delivered Smart Task Manager capstone Vanilla JS PWA with voice input, Pomodoro timer, and Service Worker offline support.

Research Assistant — AI & Computer Vision, SRMIST *Jan 2025 – Jun 2025*

Chennai, India — Hybrid

- Published IEEE paper on SAR target detection using YOLOv8 + Denoising Autoencoders, achieving 92% mAP.
- Built satellite image denoising pipelines outperforming Gaussian, Median, and Wavelet filtering (PSNR: 29.95, SSIM: 0.99).
- Processed 5,000+ SAR images using Python, PyTorch, and OpenCV. Deployed interactive demo on HuggingFace Spaces.

PROJECTS

Saarthi AI — Career Platform for Indian Students *2025*

saarthi-link.netlify.app • github.com/Balama2520/SaarthiLink React · FastAPI · ChromaDB · Saarthi AI · PostgreSQL · Tailwind CSS

- AI-powered career assistant for Indian freshers — resume ATS scoring, personalised 30/90-day learning roadmaps, job matching, and interview preparation.
- Built semantic search using ChromaDB with all-MiniLM-L6-v2 embeddings for context-aware career guidance.

Giakaa CMS — Full-Stack Content Management System *2026*

o8m-giakaa-cms.vercel.app • github.com/Balama2520/o8m-giakaa-cms React · Node.js · PostgreSQL · MongoDB · Docker · Vercel

- Architected hybrid dual-database CMS with SEO-aware rendering, OpenGraph support, XSS sanitisation, and schema validation.
- Reduced API response latency by 40% using query optimisation and efficient database indexing.

Smart Task Manager — Vanilla JS PWA *2025*

bala-smart-task-manager.netlify.app • github.com/Balama2520/smart-task-manager HTML5 · CSS3 · Vanilla JS · Web Speech API · Service Worker

- Built an offline-first PWA with task management, voice input via Web Speech API, Pomodoro timer, and time-aware gradient UI. No frameworks used.
- Features personalised name greeting via localStorage, dark/light theme, JSON export/import, and 95+ Lighthouse performance score.

AWARDS, PUBLICATIONS & CERTIFICATIONS

- **Smart India Hackathon 2024** — National Finalist (Top 5% among 10,000+ teams)
- **IEEE Published** — "Target Detection and Classification from SAR Images for Military Applications" (2025) — ieeexplore.ieee.org/document/11042
- **ResearchGate** — "Satellite Image Denoising Using Machine Learning" — researchgate.net/profile/Bala-Ayanala • huggingface.co/Balamaneesh2520
- **Certifications:** Databricks GenAI Fundamentals • AWS Cloud Fundamentals • MongoDB • Google Data Analytics